Advanced Labs Report

Boiling heat transfer experiment

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A.

1. Purpose of the experiment:

To understand the basic heat transfer characteristics of boiling, with particular focus on nucleate boiling heat transfer. In this experiment, the heating surface at the bottom of a water tank is heated by cartridge heaters. The heat flux and surface temperatures are determined from the temperature gradient inside the heated block, and the boiling characteristics are investigated.

2. Experimental Apparatus

The setup consists of a boiling heat transfer unit and a temperature measurement system:

- Water Tank: A cylindrical tank (diameter: 250 mm, height: 200 mm) containing the water to observe boiling behaviour.
- Boiling Section: A copper block where the top surface (diameter: 20 mm) acts as the heat transfer surface, surrounded by a 10 mm thick stainless-steel plate.
- Heating System: Four 1 kW cartridge heaters are press-fitted into the bottom of the copper block. Power input is regulated via a Slidac (variable transformer).
- Temperature Measurement: Four sheathed Chromel-Alumel (CA) thermocouples (ch1 to ch4) are embedded inside the copper block at depths of 4 mm, 9 mm, 15 mm, and 23 mm from the surface to measure the internal temperature gradient.

An additional thermocouple (ch6) measures the water temperature. Data is processed using a Thermodac data logger.

3. Experimental Procedure

1. Preheating: Fill the tank with water and use an immersion heater to bring the water temperature to a steady 50 ℃.
2. Initial Heating: Set the Slidac voltage to 100 V to begin heating the block.
3. Data Recording: Once the thermocouple closest to the surface (ch1) exceeds 100℃, begin recording temperatures at 1-second intervals using the data logger. Concurrently, record the boiling behavior with a digital video camera.
4. Voltage Adjustment: Increase the Slidac voltage to 150 V when ch1 exceeds $130℃Switch the temperature measurement to automatic mode when boiling behavior changes noticeably.
5. Termination & Safety: Terminate the experiment when the boiling sound becomes high-pitched and ch1 rises rapidly to 200℃ or higher, indicating dangerous onset of

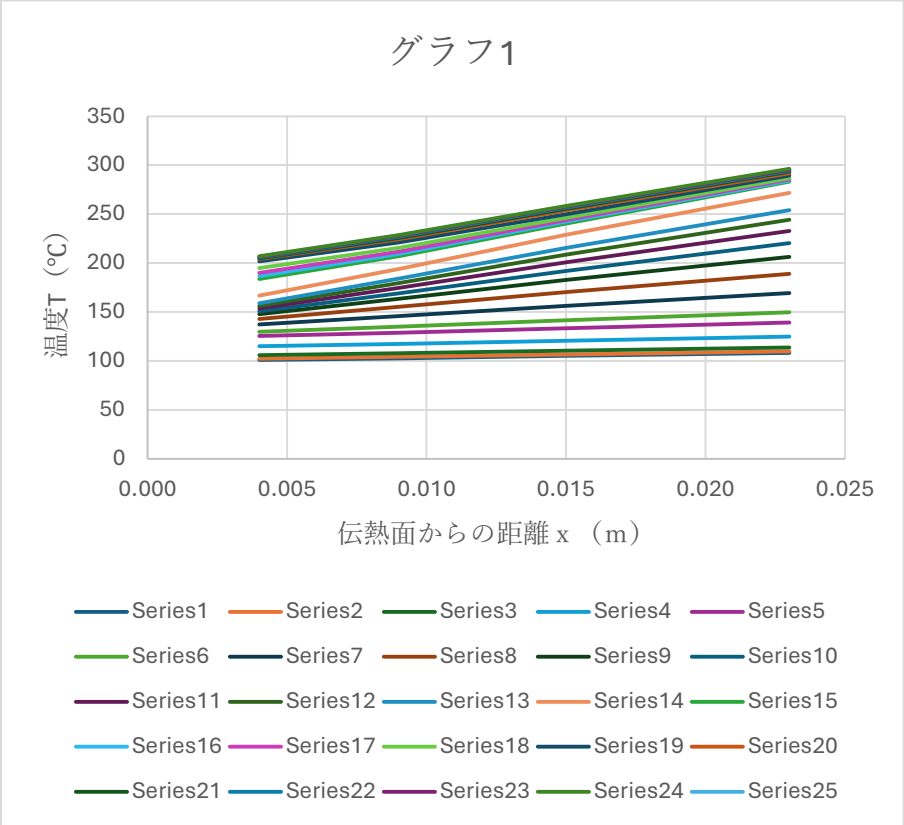
film boiling. Immediately pour cold water into the tank to lower the temperature and reduce the Slidac voltage to 0 V.

B. Arrangement of experimental data

Experimental data and graphs were recorded and made in Excel.

1. Heat flux and surface temperature

Graph 1



The relationship between the wall temperature T_w and the heat flux q(W/m^2) can be derived as the following:

The heat flux can be obtained from the temperature gradient and the thermal conductivity λ (W/(m·K)) using Fourier's law:

$$q = -\lambda \frac{dT}{dx}$$

T= temperature, x is distance

Here, using the thermal conductivity of oxygen-free copper: λ= 400 W/(m·K))

After determining q, we can then obtain the relationship between the wall temperature T_w and the heat flux q:

$q = h(T_w - T_e)$

with h= heat transfer coefficient

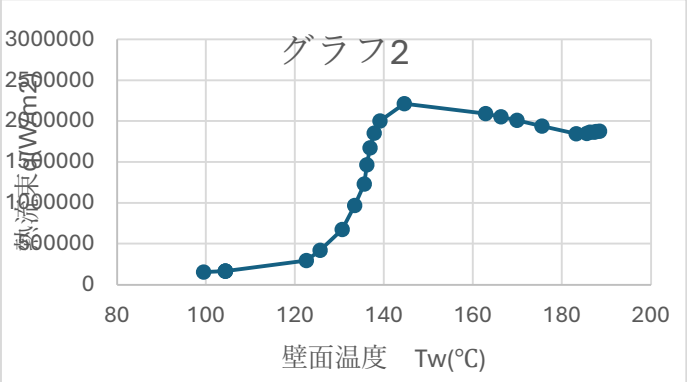
The wall temperature can then be calculated by using the equation:

$T_w = T_1 + qX_1$

where X_1 is the distance between the thermocouple and the container wall.

The relationship between heat flux and wall temperature is plotted in graph 2 below.

Graph 2



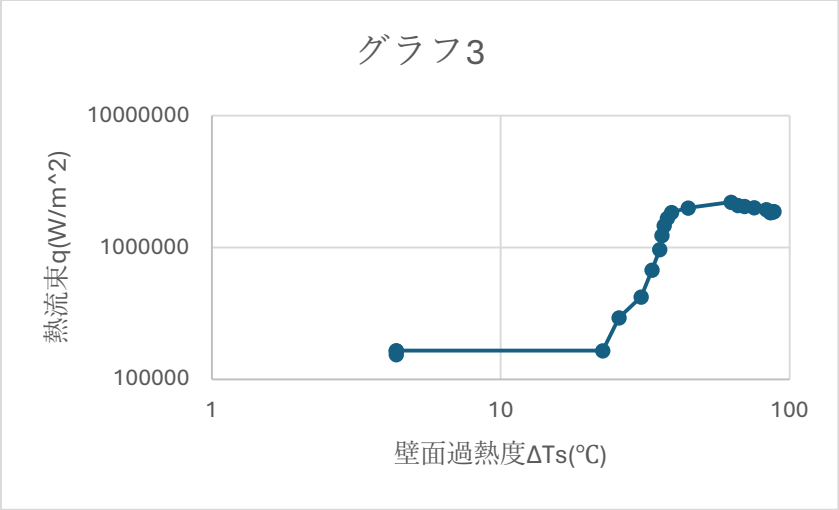
2. Heat flux and wall superheat

Graph 3

Here since the experiment is conducted at saturation temperature:

T_{sat}= 100℃ = 373 K

Relationship between heat flux and wall superheat (boiling curve) is shown in Graph 3 below plotted on log-log graph.



Based on the boiling curve in Figure 3 plotted on a log-log graph, the nucleate boiling regime started from 22℃.

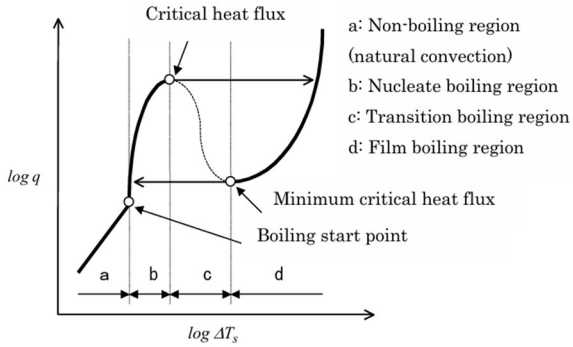


Figure 3 Boiling curve

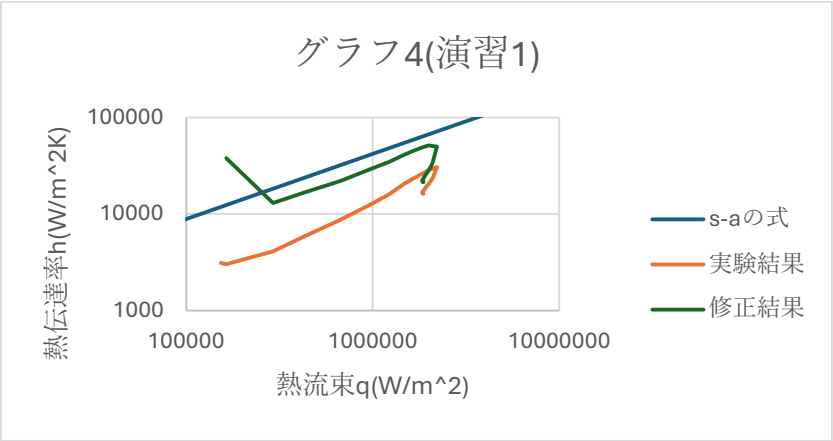
The boiling regime observed in this experiment is referred to as nucleate boiling. The difference between the saturation temperature and the bulk water temperature is called the subcooling.

3. Heat flux and heat transfer coefficient

When the water temperature is Tl , the heat transfer coefficient (h) is defined by Newtons law of cooling: q = $T_w - T_l$

The relationship between heat flux and heat transfer coefficient is plotted on a log-log graph Graph 4 below. The actual experimental data is shown on the orange

graph. Along with Stefan-Abdelsalam correlation (blue line) and corrected result (green line) which will be discussed in Exercise 1.



4. Observation of boiling temperature

Four distinct stages could be observed during the boiling process. Figure 1 shows the schematic diagram of each state.

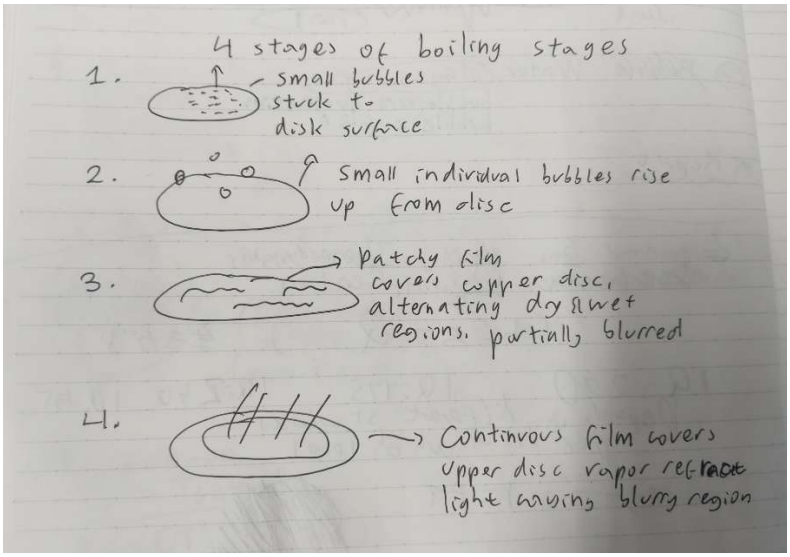


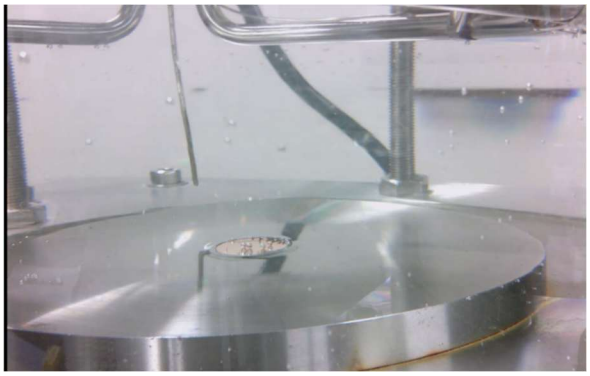
Figure 1 Schematic of boiling stage and each stage's effect on bubble formation/release

C. Discussion

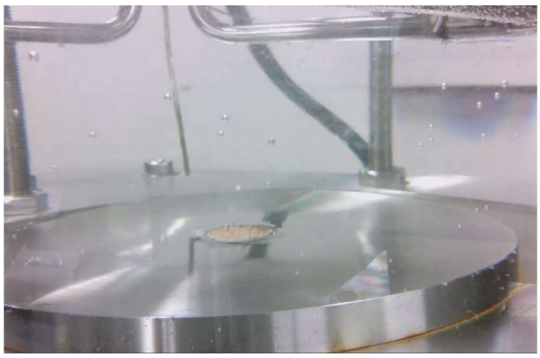
1) Comparison with experiment

Q1. Indicate which point in Fig.3 corresponds to the recorded image. Discuss the relationship between the boiling-induced flow and the heat flux at the point.

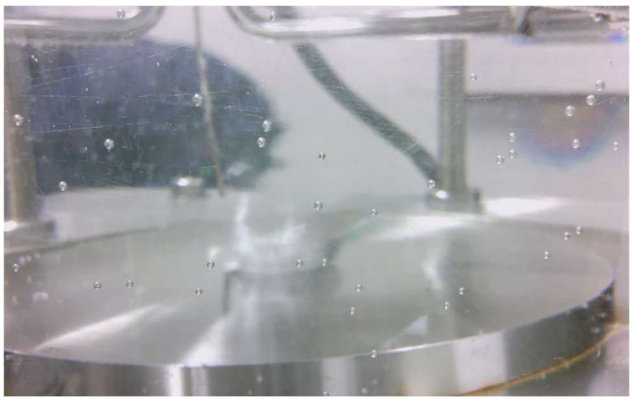
Initial bubble -> point a in Figure 3: a: Non-boiling region(natural convection)



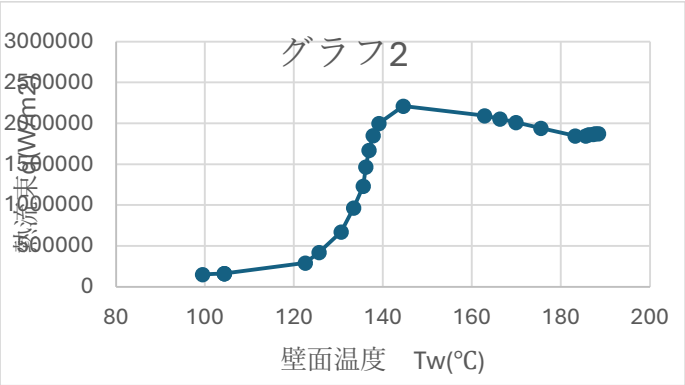
Nucleation -> point b: Nucleate boiling region



Film forming: d-> Film boiling region



Based on the change in wall temperature in Graph 2, comparing with the boiling curve in Figure 3 in the textbook, the film boiling region can be reached.



When liquid is boiled without being forced to flow, it is called pool boiling. A boiling curve is the log-log plot between the wall superheat and the heat flux such as graph 3. As heat input increases the critical heat flux point is reached. Upon further heating there is discontinuous transition from transition boiling to film boiling. Beyond this point, further heating may cause the heating surface to melt and burn out. Figure 3 in the previous pages shows the boiling curve of a typical liquid.

1. In the 1st image in Q1, which shows nonboiling region

Before boiling only small bubbles stuck to the surface are visible there is no heat flow in this stage. The bubbles are due to expansion of gasses trapped in the water.

2. In the 2nd image in Q11, which shows nucleate boiling region

Some bubbles form and heat is transferred by convection. There are few bubbles so heat transfer is slow. Most of the copper disc is covered by stationary liquid.

After the nucleate boiling region, the experiment progressed until it goes to the transition boiling region and film boiling region. This then followed by steps 3-5

3. Near the critical heat flux the boiling is most efficient. There is rapid bubble formation. Hot vapor rises up and is replaced by cold water 4. Beyond the critical heat flux bubbles form rapidly and form an insulating layer that blocks cold water from flowing down and cooling the disk. The heat flow is now choked. Causing the heat flux to lower to the minimum heat flux 5. This finally causes the copper wall to heat up rapidly as there is no liquid convection. At a high enough temperature

radiation heat flow takes over since it depends on T4 . This causes the heat flux to rise rapidly. This is shown in the 3rd image.

Q2 Discuss up to which phenomena in Fig.3 measurements could be obtained in this experiment. Discuss this based on the change in wall temperature.

The temperature measurements could only be reliably measured until the minimum critical heat flux point i.e the onset of film boiling. Beyond this point the temperature of the wall jumps 100s of degrees celsius rapidly and cannot be measured by standard thermocouples. Further the water heater and container walls may burnout and melt due to the raoid temperature increase

Q3. Discuss the reason for the sharp rise in wall temperature at the end of the experiment.

The sharp rise in wall temperature at the end of the experiment is caused by the limit in nucleate boiling which is reached at point D. In a short time, the heating surface becomes dry, and the wall temperature rises rapidly, causes a transition to point F, reaching to film boiling step which is at the end of the experiment.

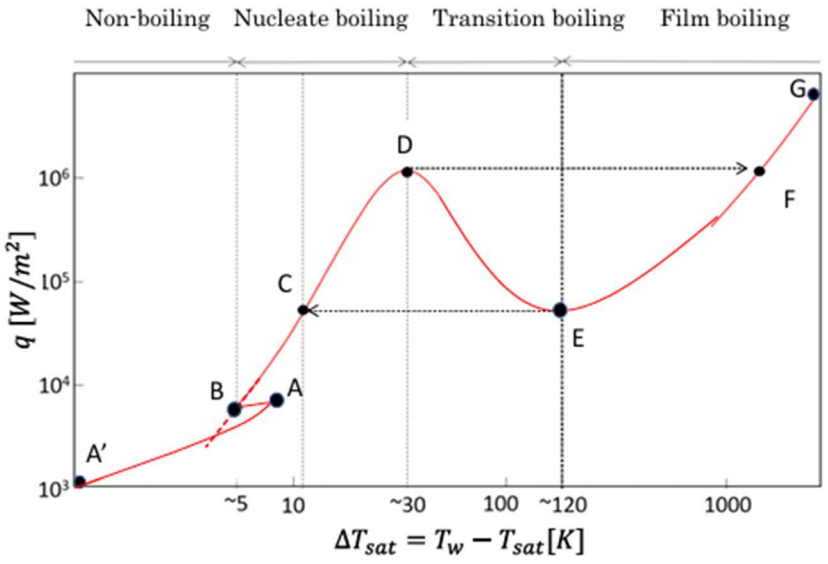


Fig.4 Boiling curvature

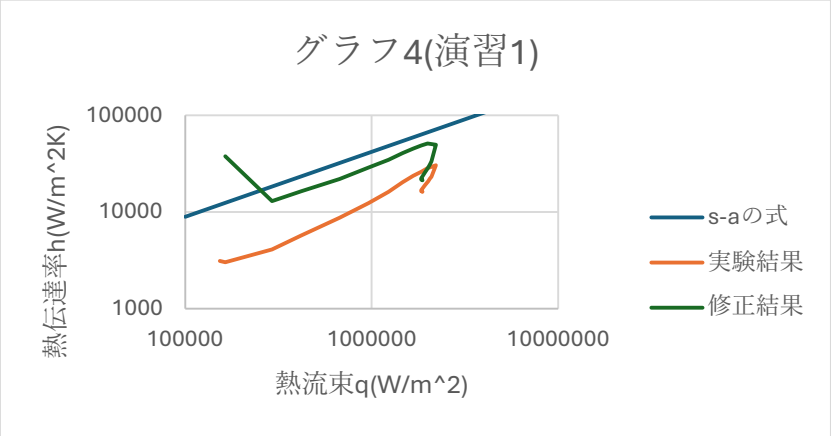
The boiling regime at point F is called film boiling, in which a stable vapor film forms between the heating surface and the liquid, and heat transfer occurs mainly through

conduction across the vapor film. Since the thermal conductivity of vapor is much lower than that of liquid, a much larger temperature is required to maintain the same heat flux. As the result, the wall temperature increases sharply under heat-flux-controlled setting.

From the temperature history of the heating surface which are shown in Graph 3, we can see that the rapid rise in wall temperature is reached at around 120℃, at which point D (limit of nucleate boiling) occurs. The temperature of the water was increased beyond 130℃ until it reached around 180℃ at the end of the experiment at which the onset of the film boiling occurs.

2) Comparison with a conventional empirical correlation [Exercise 1]

Graph 4 shows experimental results of the relationship between heat flux q and heat transfer coefficient h overlayed with results calculated by the Stephen-Abdelsalam correlation, and the corrected results.



To obtain the results in Graph 4 in Exercise 1 , do the following

(i)Plot the relationship between heat flux "and heat transfer coefficient *h* on a log–log graph over the specified range according to the S-A equation. Recalculate h, replacing Ti (the thermocouple measurements) with the saturation temperature Tsat , and plot the experimental correlation once again.

The Stefan-Abdelsalam correlation is simplified and from the textbook is adapted to calculate heat transfer coefficient.

$$h = 3.84 \times q^{0.673}$$

Q4 Compare the Stephan–Abdelsalam correlation, the experimental results, and the corrected results. Discuss the reasons for any differences and comment on the applicability of the Stephan Abdelsalam correlation to this experiment.

The Stephan-Abdelsalam correlation shows a linear relationship between heat flux and heat transfer coefficient, whereas the experimental results and the corrected results are nonlinear. Furthermore, the Stephan-Abdesalam correlation lies above both the experimental results and the corrected results. All of the experimental results are below and do not intersect with the Stephen-Abdelsalam correlation. The corrected results, on the other hand, move closer to and partly intersect with the Stephen-Abdelsalam correlation. This implies that the correlation is compensating for an unmodeled effect in the original correlation.

To further elaborate, 4 key observations can be made:

1. The linear region of revised fits to S-A equation reasonably well

2. The unrevised experimental data predicts much lower q than S-A equation and revised.

3. There is a flat region in the revised results

4. Near the critical heat flux a hook is observed

The curves differ to some of the following physical and experimental factors:

1. Surface condition / heater morphology. Stephan–Abdelsalam is an empirical correlation built from other data, so it is sensitive to surface finish, roughness, and nucleation-site density. Differences in polishing, oxidation, contamination, or microscopic cavities can shift the boiling performance substantially.
2. Property and fluid mismatch. The correlation works best when the fluid and pressure conditions are close to the database used to fit it. If your fluid, pressure, subcooling, or heater material differ, the predicted heat transfer coefficient can be systematically high or low.
   Even though the Stefan-Abdelsalam correlation used in this experiment may be based on same fluid(water) and under the condition of ΔT_{sub}=0 and the average error is 11.3% in the range of $10^{-4} < \frac{P}{P_c} < 0.886$, the simplified formula to determine the heat transfer h based on the Stefan-Abdelsalam correlation in the textbook
   $$h = 3.84 \times q^{0.673}$$

may not fully reflect real conditions during the actual experiment.

3. Low-heat-flux regime effects. At low heat flux, natural/free convection contributes a lot, so pure nucleate-boiling correlations often show larger error there.
4. Two-phase dynamics at higher heat flux. Bubble interaction, coalescence, and departure behaviour become important as heat flux rises, and these effects are not fully captured by a simple correlation.
5. Measured value of Ti Lower than Tsat. The experimental results before revision form a much lower line than the S-A equation and

   $$h = \frac{q}{T_w - T_i}$$

   The evaluated heat flux was much lower than the true value. The explanation for this is that the themocouple measured the bulk water temperature which was much lower than the temperature at the boiling surface.
6. The flat region may have arisen because boiling had not begun yet and there is no increase in heat flux.The S-A model assumes that the liquid is already saturated and undergoing nucleate boiling.
7. Behaviour near critical heat flux
   Near the critical heat flux point transition boiling occurs. In this region the heat flux may rapidly oscillate since there is alternation between insulating film formation and convective heat transfer due to bubble formation. This is the result for the hooks in the ends of the experimental graphs.
8. Other experimental errors.
   Measurement errors in the thermocouple, variations in external conditions such as pressure may cause differences in the predicted vs experimental boiling behaviour. This could explain the slightly lower revised experimental measurements compared to the S-A equation

The corrected result appears to be a fitted or adjusted version of the correlation, likely including a correction factor for our specific heater or liquid. In the literature, Stephan–Abdelsalam is often described as a good general correlation, but some studies still apply tuning factors because no single boiling correlation works well for all surfaces and fluids. That is consistent with the plot: the correction brings the prediction closer to the measured data by accounting for experiment-specific behaviour that the original equation misses.

For the experiment, Stephan–Abdelsalam is qualitatively applicable but not fully predictive without correction. It seems useful as a first estimate and for checking whether the boiling regime is reasonable, but the raw correlation overpredicts or underpredicts the actual data enough that it should not be used as-is for final design or quantitative interpretation. If our goal is accurate heat-transfer prediction for this specific setup, the corrected curve is the more appropriate one to use, while Stephen-Abdelsalam correlation (s-a 式) in the graph should be treated as a baseline reference rather than a final model.

The Stephan–Abdelsalam correlation reproduces the overall monotonic dependence of heat transfer coefficient on heat flux, but the absolute values differ from the experiment because the correlation does not fully account for the heater surface condition and experiment-specific boiling behavior. The corrected result improves agreement by compensating for these effects, so the uncorrected Stephan–Abdelsalam correlation is only partially applicable to this experiment.

Other applicabilities of the Stefan Abdelsalam correlation:

- The approximate fit between the revised results show that the S-A equation is applicable to our experiment within the linear region.
- We must consider the water to be at the saturation temperature.
- The S-A equation is only applicable in the nucleate boiling region, and cannot be used past the critical flux point.
- • The water must initially be at Tsat to match the result of the S-A equation.